

David Sebastián Rodríguez Numpaque

Scientific Software Developer

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About Me

BSc in Astronomy focused on exoplanetary science and computational astrophysics. Experienced in scientific software development, spectral analysis, photometric modeling, collaborative research, and academic outreach. Lead developer of a Bayesian, cluster-parallelized inference pipeline for exoplanetary ring detection, and co-developer of an open-source simulation package. Additional interests include machine learning, AI applied to science, and high-energy astrophysics.

Skills

Programming & Tools: Python (Scientific computing, OOP), C, Bash & Linux, Git/GitHub, Parallel computing, Cluster/HPC, Sphinx/RTD, Jupyter, LaTeX	(clustering), Deep learning fundamentals
Bayesian & Statistical Computing: Bayesian inference, Nested Sampling, MCMC, Forward Modeling	Data Access & Databases: SQL, Relational databases, ADQL (Virtual Observatory), FITS file handling
Machine Learning & Data Science: scikit-learn, Supervised learning (classification, SVM), Unsupervised learning	Scientific Python Packages: NumPy, SciPy, Pandas, Matplotlib, Seaborn, Plotly, Astropy, Astroquery
	Astronomy Software: IRAF, iSpec, Rebound, Lightkurve, Batman, pyPplusS, Pryngles, Spiderman

Technical Experience

Bayesian Inference Pipelines

- Designed and developed PRisma, a modular Bayesian inference framework for exoplanetary research, integrating forward modeling, KDE-based likelihood estimation, and nested sampling (dynesty). Implemented parallel execution on HPC clusters, reproducible workflows, and version-controlled inference campaigns, enabling large-scale analyses and supporting ongoing first-author research.

Python, Bayesian inference, Nested sampling (dynesty), Parallel computing, HPC, Scientific computing, Reproducible research

Applied Data Mining & Machine Learning

- Applied end-to-end data science workflows to astronomical datasets, including SQL/ADQL querying, data preprocessing, visualization, feature engineering, and supervised and unsupervised machine learning. Developed predictive and clustering models using the Python scientific ecosystem as part of research-oriented projects.

SQL, ADQL, scikit-learn, Data analysis, Classification, Clustering, Machine learning, Data visualization

Scientific Software Engineering

- Co-developed and maintained Pryngles, an open-source Python package for photometric simulations of planetary systems. Designed and evolved its public API, implemented testing and documentation infrastructure, and applied modern software engineering practices including modular architecture, collaborative Git workflows, reproducible development, and long-term maintainability.

Python, Software architecture, API design, Open-source development, Git / GitHub, Unit testing, Documentation, Reproducible research

Publications

- Numpaque, S.**, Zuluaga, J. I., Alvarado-Montes, J. A., & Kipping, D. (2026). Probing Exoplanetary Rings with Asterodensity Profiling: A PhotoRing Analysis of Kepler-51. (*In preparation*)

2. Valderrama, D. A., **Numpaque, D. S.**, et al. (2024). Artificial Intelligence in the Teaching of Natural Sciences on the Threshold of the Fifth Industrial Revolution. *Explainable AI for Education: Recent Trends and Challenges* – book chapter. https://doi.org/10.1007/978-3-031-72410-7_9

Education

BSc in Astronomy — University of Antioquia, Medellín, Colombia (2021 – *Expected graduation: June 2026*)

- Outstanding Advanced Student in Astronomy (2025)

Complementary Courses

- Research Project Design | Office of Research, University of Antioquia, 2025 – *Certified*
- The Diversity of Exoplanets | University of Geneva, 2024 – *Coursera Certified*
- Data Mining | Pontifical Catholic University of Chile, 2021 – *Coursera Certified*
- Introduction to AI | DeepLearning.AI, 2021 – *Coursera Certified*

Outreach & Education

Organizational Manager — Astrodidaxi Foundation, Colombia (2023 – Present)

- Coordinated outreach initiatives, research projects, and public engagement activities to strengthen astronomy education in Colombia.
- Organizing Committee member for the II & III Workshop on Astronomy Education (OAE-Colombia, Astrodidaxi, UPTC).

Mentorship Program Organizer – RECA, Colombian Astronomy Students Network (2025)

- Coordinated the national call for applications, organized panels, and managed logistics and communication for the RECA Mentorship Program.

Co-founder & Coordinator – Astronomy Research Seedbed, University of Antioquia (2025)

- Designed and led a training program for undergraduate students focused on computational astrophysics, research methods, and scientific communication.

Conferences

Speaker

- **XXI National Astronomy Meeting, RAC 2025, Colombia**
Strengthening Research Training in Astronomy: The First Astronomy Research Seedbed at the University of Antioquia
- **VIII Colombian Congress of Astronomy and Astrophysics (CoCoA) 2024, Colombia**
A Python Package for Modelling the Photometric Signatures of Exoplanetary Rings and Satellites
- **III International Congress on Community Environmental Education (CIEAC) 2024, Colombia**
Exploring the Contribution of Astronomy to the Sustainable Development Goals and Community Empowerment in Boyacá

Assistant

- II Ibero-American Congress of Physics Students (CIEF), Peru — 2023
- VII Colombian Congress of Astronomy and Astrophysics (CoCoA), Colombia — 2022

Languages

- Spanish (Native) · English (B2)